# Ross Edwards

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## Profile

Power Analyst with 2 years of European power market experience, focused on CWE. Quantitative background in stack modelling, optimisation, and demand forecasting, with ownership of in-house fundamental and pricing models. Looking to continue my career in quantamental power analysis and trading outside the UAE.

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## Relevant Experience

**Cobblestone Energy** Dubai

*Power Analyst* Oct 2024 – Present

- Independently built, productionised, and deployed the in-house stack model and central short-term power trading application, incorporating Core/Nordic flow-based market coupling, MILP for thermal commitment, PSH/BESS optimisation, backtesting framework, custom scenarios, and explainable restricted LP price formation.
- Developed in-house demand models from scratch using an ensemble of boosted trees, LSTM, and N-HiTS, outperforming vendor forecasts across every European bidding zone, with $> 25\%$ lower 2026 YTD WMAPE across all key target markets in CWE, Hungary, and Italy.
- Developing ML unit-level availability forecasts, including outage return modelling and weather-driven effects, to improve inputs into downstream pricing models.
- Actively trade CWE day-ahead to front-month power in my own book, while also publishing daily powerburn views for the gas trading team.
- Ranked 1st of 5 in my graduate cohort by PnL and risk progression during an initial GB intraday power trading desk rotation.

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## Selected Projects & Achievements

**JEPX Power Pricing Model**

- Built a short-term Japanese power stack model across all 9 regions, ingesting data from OCCTO, JEPX, HJKS, and the EPCOs to generate zonal price forecasts.
- Developing a hybrid pricing model combining deep learning and fundamental merit-order price forecasts, investigating how the relative value of each approach changes across market conditions.

**Agent-Based Modelling of Collective Cell Dynamics**

- Developed a C++ agent-based simulation of collective cell behaviour in tissue-like environments relevant to wound healing, tumour invasion, and early embryogenesis.
- Analysed emergent tissue dynamics including collective migration, tissue fingering, and disassembly to identify biological mechanisms relevant to drug target discovery.

**Competitive Gaming**

- Ranked #1 globally in FIFA against AI and top 100 globally against human players, out of 20 million.

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## Technical Skills

- **Programming:** Python (production), SQL, C++ (projects)
- **Power Markets:** Stack modelling, flow-based market coupling, demand, storage, availability forecasting
- **Modelling Areas:** Optimisation, time-series, anomaly detection, classification, agent-based modelling
- **Machine Learning:** XGBoost, LightGBM, CatBoost, LSTM, N-HiTS, MLP, Logistic Regression
- **Deployment:** AWS (EC2, EKS), Docker, Kubernetes

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## Education

**Durham University** Sep 2019 – Jul 2023

- *MSci Natural Sciences (Physics & Biology), 2:1*
- *Specialisation: Computational Biophysics*
- Entered university one year early following exceptional academic performance at school.